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Blockchain

References:
1. Bitcoin and Cryptocurrency Technologies
Arvind Narayanan et al.
2. Lectures on Blockchain Technology by Prof. Sandeep Shukla, IIT Kanpur
3. Bitcoin: A Peer-to-Peer Electronic Cash System by Satoshi Nakamoto

Recap

- PAXOS
- DSM
  - Strict consistency / atomic consistency / linearizability
  - Sequential Consistency

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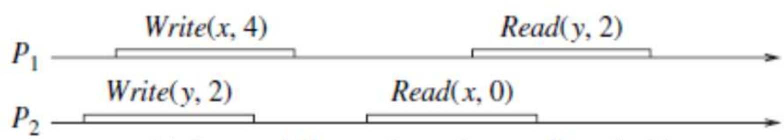
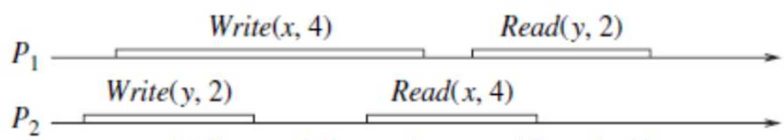
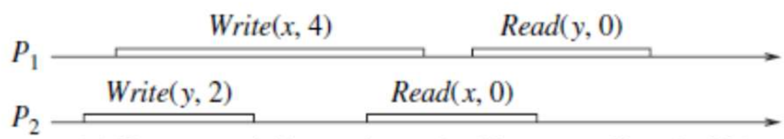
Consistency



(a) Sequentially consistent but not linearizable



(b) Sequentially consistent and linearizable



(c) Not sequentially consistent (and hence not linearizable)

Consistency

Linearizability

- guarantee about single operations on single objects.
- provides a real-time guarantee on the behavior of a set of single operations on a single object

Serializability

- guarantee about transactions (groups of one or more operations over one or more objects)
- guarantees that the execution of a set of transactions over multiple items is equivalent to some serial execution of the transactions.

- A consistency model that is weaker than strict consistency but stronger than sequential consistency is linearizability (?)

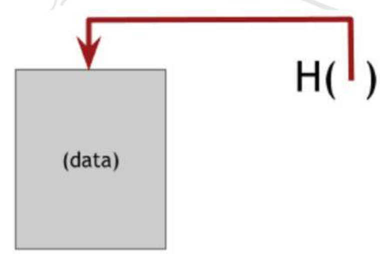
  \- Tannenbaum

Blockchain

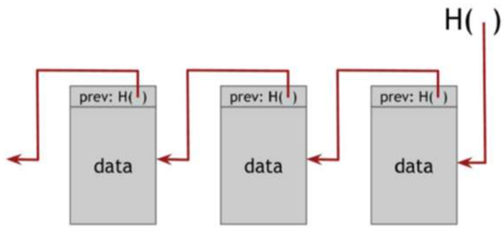
- Blockchain was introduced with the invention of Bitcoin in 2008
- It is a distributed and replicated linked list of blocks
- But this linking is not through memory addresses
- Cryptographic technique called hashing is used for linking
- Consistency of the data is maintained by a process called consensus.
- **Consensus**: everybody (or majority) agrees that the data that goes into the data structure is what they agree to put there

Hash Pointer

- A hash pointer is a pointer to where some information is stored together with a cryptographic hash of the information.
- A regular pointer gives you a way to retrieve the information
- A hash pointer also gives you a way to verify that the information hasn't changed.



Block Chain



A linked list using hash pointers

- In a regular linked list each block has data & a pointer to the previous block in the list
- In a block chain the previous block pointer is replaced with a hash pointer.
- So each block tells us: (1) where the value of the previous block is + (2) contains a digest of that value that allows us to verify that the value hasn't changed.
- Head of the list, is a hash-pointer that points to the most recent data block.

Blockchain

- Blockchain is a log that cannot be changed by a malicious party or by mistake (It maintains integrity of the data)

  => Once all parties agree about the data to put in the

  data structure, it can not be tampered with
- The kind of computation power one needs to tamper with the log is virtually impossible to get
- One of the first application of blockchain was bitcoin
- But blockchain can be used in many other applications

Banking

- In a bank banking system, we have multiple players

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Banking

- How to do a transaction – write cheque or use internet / ATM
- When you transfer money to someone, bank performs certain checks
  - You are authorized to perform this transaction
  - Your balance is greater than your transaction amount.
- Finally Tx is accepted or rejected and action is taken
- How do the customer knows that this is done properly and correctly?
  - Check online or from monthly statement
- Who maintains the ledger from which this monthly statement is generated?

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Banking

- & because of regulatory framework that we have, banks do not do something wrong.
  - Tx done almost always appear in the statement
  - Tx not done do not appear in the statement.
- What if bank allows an invalid Tx and one complains
  - It was not authorized (someone stole password)

    (prove it - non repudiation)
  - Balance was not sufficient (or someone added zeros in the amount written by you) (integrity)
- We trust banks because they are answerable to the regulatory authority RBI - trusted third party enforced trust

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Blockchain

- The blockchain could be one of the solutions
  - It can provide transparency and can maintain the integrity of the data and
  - can provide trust without a trusted third party

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E-cash

- Two fundamental e-cash system issues need to be addressed:
- **Accountability – (1)** cash is spendable only once (double-spend problem) and **(2)** it can only be spent by its rightful owner.
- As it is quite easy to make copies of digital data, this becomes a big issue in digital currencies as you can make many copies of same digital cash.
- **Anonymity** is required to protect users' privacy.
- As with physical cash, it is almost impossible to trace back spending to the individual who actually paid the money.

E-cash

- David Chaum solved both of these problems during his work in 1980s by using two cryptographic operations,
  - **Blind signatures** and
  - **Secret sharing**

Blind Signatures

- One of the simplest blind signature schemes is based on RSA signing.
- A traditional RSA signature (e, N) (d, N)
- The blind version uses a random value r (r is relatively prime to N).
- *r* is raised to the public exponent *e* modulo *N*, and the resulting value is used as a blinding factor.
- The author of the message computes the product of the message and blinding factor, and sends the resulting value m' to the signing authority.

$$m' \equiv mr^e \pmod N$$

Blind Signatures

- The signing authority then calculates the blinded signature *s'* as:

$$s' \equiv (m')^d \pmod N \quad (d, N)$$

- *s'* is sent back to the author of the message, who can then remove the blinding factor to reveal *s*, the valid RSA signature of *m*

$$s \equiv s' \cdot r^{-1} \pmod N \quad (m')^d \cdot r^{-1}$$

- This works because RSA keys satisfy the equation $(mr^e)^d r^{-1}$

$$r^{ed} \equiv r \pmod N \quad m^d \underbrace{r^{ed}}_{r} r^{-1}$$

- hence *s* is indeed the signature of *m*.

(ref: Wikipedia)

Hash Function

- A ***hash function*** is a mathematical function with the following three properties:
  - Its **input** can be any string of any size.
  - It produces a fixed size **output** (assume a 256-bit output size)
  - It is efficiently computable. (More technically, computing the hash of an *n*-bit string should have a running time that is O(*n*)).

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Cryptographic hash functions

- For a hash function to be cryptographically secure, we require that it has the following additional properties:

  (1) collision-resistance, $y = H(x)$

  (2) Hiding (pre-image resistance) $H(x) = H(y)$

  (3) Second Pre-image resistance

  (3) puzzle-friendliness
- **Collision-resistance:** A hash function *H* is said to be collision resistant if it is infeasible to find two values, *x* and *y* , such that $x \neq y$, yet $H(x) = H(y)$

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Cryptographic hash functions

- ***Hiding*** This property asserts that if we're given the output of the hash function $y = H(x)$, there's no feasible way to figure out what the input, x , was.

- **Second Pre-image resistance:** Given an input x, it's computationally infeasible to find another input x' such that h(x) = h(x')

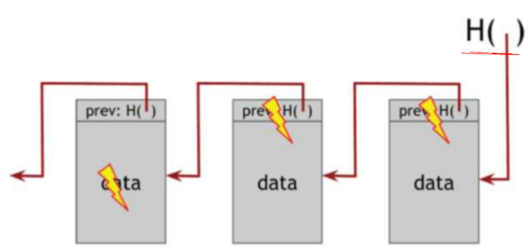
2^{256}

$H(m \| nonce) =$

- **Puzzle friendliness.** A hash function H is said to be puzzle-friendly if for every possible n-bit (target) output value y , if k is chosen from a distribution with high min-entropy, then it is infeasible to find x such that H(k ‖ x) = y in time significantly less than 2^n

Block Chain

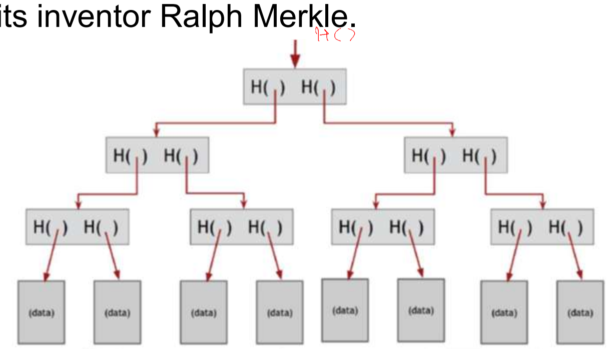
- A block chain is a **tamper-evident log**



- The very first block of the block chain - **genesis block**

Merkle trees

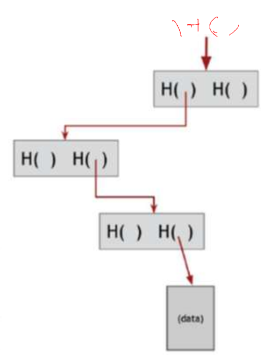
- A binary tree with hash pointers is known as a ***Merkle tree*** , after its inventor Ralph Merkle.



- we need to remember just the hash pointer at the head of the tree
- Designed to separate transaction data from its cryptographic proof

Proof of membership

- We remember just the root
- What is needed to prove that a certain data block is a member of the Merkle Tree?
- One need to show only this data block, and the blocks on the path from the data block to the root.
- If there are n nodes in the tree, only about log(n) items need to be shown
- It takes about log(n) time for us to verify it.



Merkle trees

- if an adversary tampers with some data block at the bottom of the tree, the change will eventually propagate to the top of the tree
- Which can't be tampered